+1 (778) 991-3647
Vancouver, BC
emirozarda@hotmail.com

Emir Ozarda

Mechanical Engineer-in-Training

Design Portfolio
LinkedIn

Skills

Tools and Software	SolidWorks, AutoCAD, Siemens NX, Inventor, Geomagic Design X, SAP BEAS, SolidWorks ePDM, DBWorks, ANSYS, JobBOSS, DraftSight, Fusion 360, Catia, VeriCut, Excel, Hilti PROFIS Anchor
Programming Languages	R, C, Assembly, MATLAB, LaTeX
Communication	English, French, Turkish
Hands-on	FARO Arm, 3D Scanner, Manual Measurement, 3D Printer, Laser Cutter, Waterjet, 3&5–Axis Milling Machine
Certifications	CSWP-Mechanical Design, Intermediate First Aid, CAD and Digital Manufacturing

Education

University of British Columbia, *Bachelor of Applied Science, Mechanical Engineering Major,* ***GPA:*** *3.2/4.0*
Awards: *Faculty of Applied Science International Student Scholarship*

Work Experience

Mechanical Designer — **June 2024 — Present**
SCG Process — *Langley, BC*

- **Reverse engineered** pump components from numerous market-available pumps, transforming worn samples into detailed **production drawings**.
- Employed **3D scanning**, **FARO Arm**, and **manual measurements** techniques to capture precise geometry from pump parts, then extracted data in **Geomagic Design X** to rebuild accurate SolidWorks models with machine allowances and as-cast configurations.
- Dimensioned and toleranced individual parts to achieve precise fits up to 0.0001 inches, ensuring proper running clearances for pump assemblies.
- Incorporated **O-ring grooves**, **surface finishes**, **bearing fits**, and other critical features into released production drawings in accordance with industry and company standards.
- Standardized components in collaboration with the production team, developing and implementing **ECOs** for released drawings.

Mechanical Designer — **November 2022 — April 2024**
Love The Drive — *Richmond, BC*

- Designed steel brackets for wind deflectors using **SolidWorks Sheet Metal**, manufactured and validated prototypes with **3D FDM printing**.
- Analyzed failure modes and gathered production-team input to improve existing production jigs.
- Developed a new product by benchmarking market offerings and refining components to reduce new design and purchasing costs.
- Managed products from design through manufacture, ensuring accurate assembly and testing of finished products.
- Researched **low-** and **high-carbon steel**, calculated minimum thickness for bending loads, and estimated manufacturing cost to select the most cost-effective option.

Mechanical Engineering Co-op — **January 2023 — August 2023**
WhiteWater West Industries — *Richmond, BC*

- Redesigned composite parts with reinforcement and post-demolding features such as **coring**, **ribs** and **bolt holes**.
- Used SolidWorks **welding** package to design guardrails with structural members, creating preliminary **BOMs** and **cut lists**.
- Created and reviewed detailed **production drawings** with accurate **GD&T**, including laminate schedules, dam information, trim perimeters, and bolt-hole locations as a part of the drafting team.
- Developed a test plan referencing ICBC 2015 Load Test criteria to verify FRP part compliance with design codes.
- Managed project data using **PDM** software and facilitated the transition from legacy to current PDM systems.
- Prepared comprehensive reports on blister, adhesive-scratch, flexural, and tensile tests using UTM, strain gauge, and extensometer.

Mechanical Engineering Co-op — **May 2021 — September 2021**
Transvaro Electron Instruments Ind. Trade Co. — *Istanbul, Turkey*

- Designed an aluminum component for **air leak testing** on a multi-vision lens compartment, from concept through **structural analysis** and safety-factor refinement.
- Selected **Aluminum 6061** for machinability, low-volume suitability, and cost, based on comparative material research.
- Created detailed technical drawings with accurate **GD&T**, and generated comprehensive **BOMs** with Parent-Child Relationships.
- Manufactured part on a **3-axis CNC Mill**, generating toolpaths in **Siemens NX CAM**.

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Milestone Projects

Quadcopter Drone Frame **June 2022 — January 2023**

- Designed a quadcopter frame in **Autodesk Fusion 360** using **Generative Design** to minimize mass while maintaining load capacity, reducing body weight by **370 grams** and achieved 4/1 thrust ratio.
- Conducted **Static Stress Analysis** to identify stress concentrations and verify the safety factor, producing both a 3D-printed (nylon 6) and CNC-machined aluminum alternative, generating toolpaths for **waterjet manufacturing**.

Smart Carrier **September 2021 — April 2022**

- Reduced Torque and RPM requirements by **75 percent** via **SolidWorks Motion Study** analysis across wheel and frame configurations for a self-driving robot.
- Designed a **drivetrain** from scratch, selecting gear ratios and a **gearmotor** to meet torque-speed targets, and built a chassis rated to 500N loads using wood and aluminum.